

Introduction to Calculus

Calculus is the mathematics of change and accumulation. It has two main branches: differential calculus, which studies how quantities change instantaneously (rates of change, slopes of curves), and integral calculus, which studies the total accumulation of quantities (areas under curves, total distance traveled). The two branches are deeply connected by the Fundamental Theorem of Calculus, which shows that differentiation and integration are essentially inverse processes. Understanding calculus begins with the idea of a limit, which lets us talk about values that a function approaches as the input gets closer and closer to some point.

Limits – The Foundation

A limit describes the behavior of a function as its input approaches a particular value, even if the function is not defined at that exact point. We write $\lim_{x \rightarrow a} f(x) = L$ to mean “as x gets arbitrarily close to a , the value of $f(x)$ gets arbitrarily close to L .” For example, consider $f(x) = \frac{x^2 - 1}{x - 1}$. At $x = 1$ the formula gives $\frac{0}{0}$, which is undefined, but if we simplify $f(x) = x + 1$ for $x \neq 1$, we see that as $x \rightarrow 1$, $f(x) \rightarrow 2$. Hence $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2$. Limits allow us to define continuity, derivatives, and integrals rigorously.

Continuity – When a Function Has No Breaks

A function f is continuous at a point $x=a$ if three conditions hold: (1) $f(a)$ is defined, (2) $\lim_{x \rightarrow a} f(x)$ exists, and (3) the limit equals the function value: $\lim_{x \rightarrow a} f(x) = f(a)$. Intuitively, you can draw the graph of a continuous function without lifting your pen. Continuity is essential because the derivative of a function at a point exists only if the function is continuous there (though continuity alone does not guarantee differentiability). For instance, $f(x)=|x|$ is continuous everywhere but has a sharp corner at $x=0$, so it is not differentiable there.

The Derivative – Instantaneous Rate of Change

The derivative of a function f at a point $x=a$ measures the slope of the tangent line to the graph at that point. It is defined as the limit of the average rate of change over an interval as the interval shrinks to zero:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

If this limit exists, we say f is differentiable at a .

Worked example: Find the derivative of $f(x) = x^2$ at any x .

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\ &= \lim_{h \rightarrow 0} (2x + h) = 2x. \end{aligned}$$

Basic Differentiation Rules

Once we know the definition, we can differentiate many functions quickly using rules derived from it.

- Constant rule: $\frac{d}{dx}[c]=0$.
- Power rule: $\frac{d}{dx}[x^n]=n x^{n-1}$ for any real n .
- Sum rule: $\frac{d}{dx}[f(x)+g(x)]=f'(x)+g'(x)$.
- Product rule: $\frac{d}{dx}[f(x)g(x)]=f'(x)g(x)+f(x)g'(x)$.
- Quotient rule: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]=\frac{f'(x)g(x)-f(x)g'(x)}{[g(x)]^2}$.
- Chain rule: $\frac{d}{dx}[f(g(x))]=f'(g(x)) \cdot g'(x)$.

Worked example: Differentiate $h(x)=(3x^2+5)^4$.

Let $u=3x^2+5$, so $h=u^4$. Then

$$\frac{dh}{dx}=4u^3 \cdot \frac{du}{dx}=4(3x^2+5)^3 \cdot (6x)=24x(3x^2+5)^3.$$

Applications of the Derivative – Optimization

Derivatives help us find where a function increases or decreases and locate its maximum or minimum values. If $f'(x) > 0$ on an interval, f is increasing there; if $f'(x) < 0$, it is decreasing. A point where $f'(x) = 0$ (or where f' does not exist) is a critical point; it may be a local maximum, minimum, or neither. The second derivative test can often decide: if $f''(c) > 0$ then f has a local minimum at c ; if $f''(c) < 0$ a local maximum.

Worked example: A rectangle has a fixed perimeter of 20 m. Find the dimensions that give the largest area.

Let the sides be x and y . Perimeter: $2x + 2y = 20 \Rightarrow y = 10 - x$. Area: $A = xy = x(10 - x) = 10x - x^2$.

Differentiate: $A'(x) = 10 - 2x$. Set to zero: $10 - 2x = 0 \Rightarrow x = 5$. Then $y = 10 - 5 = 5$.

Second derivative: $A''(x) = -2 < 0$, so this critical point gives a maximum. The rectangle of maximum area is a square $5 \text{ m} \times 5 \text{ m}$ with area 25 m^2 .

Antiderivatives and Indefinite Integrals

If $(F'(x)=f(x))$, we call (F) an antiderivative of (f) . The process of finding antiderivatives is called integration. Because adding a constant does not change the derivative, the most general antiderivative includes an arbitrary constant (C) :

$$\int f(x)dx = F(x)+C.$$

The power rule for integration (the reverse of the power rule for differentiation) states:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1).$$

Worked example: Find $\int (4x^3-2x+7)dx$.

Integrate term by term:

$$\int$$

The Definite Integral and the Fundamental Theorem of Calculus

The definite integral $\int_a^b f(x) dx$ represents the signed area between the curve $y=f(x)$ and the x -axis from $x=a$ to $x=b$. It can be approximated by Riemann sums (thin rectangles) and defined as the limit of those sums as the width goes to zero.

The Fundamental Theorem of Calculus links differentiation and integration:

Part 1: If F is an antiderivative of f on $[a,b]$, then $\int_a^b f(x) dx = F(b)-F(a)$.

Part 2: If f is continuous, then the function $g(x)=\int_a^x f(t) dt$ satisfies $g'(x)=f(x)$.

Worked example: Evaluate $\int_0^2 (3x^2) dx$.

An antiderivative of $(3x^2)$ is (x^3) (since $\frac{d}{dx}x^3=3x^2$).

Apply Part 1:

[

Common Mistakes in Calculus

1. Misapplying the chain rule: Forgetting to multiply by the derivative of the inner function (e.g., differentiating $\sin(x^2)$ as $\cos(x^2)$ instead of $2x\cos(x^2)$).
2. Ignoring the constant of integration: When computing an indefinite integral, omitting $+C$ loses the family of all antiderivatives.
3. Confusing derivative and integral rules: Thinking that $\int f(x)g(x)dx = (\int f(x)dx)(\int g(x)dx)$; integration does not distribute over products like differentiation does.
4. Assuming continuity implies differentiability: A function can be continuous but not differentiable at a corner or cusp (e.g., $|x|$ at $x=0$).
5. Evaluating limits by direct substitution without checking: If substitution yields $\frac{0}{0}$ or $\frac{\infty}{\infty}$, further work (factoring, rationalizing, L'Hôpital's rule) is needed.

Practice Problems

Try these on your own. Answers are hidden—check your work after you attempt them.

1. Differentiate $f(x) = \frac{(2x+1)^3}{x^2+4}$.
2. Evaluate $\int_1^3 (5x^2 - 2x + 1) dx$.

(When you have solved them, you can reveal the answers:)

<details><summary>Show answers</summary>

1. $f'(x) = \frac{6(2x+1)^2(x^2+4) - 2x(2x+1)^3}{(x^2+4)^2}$ (you may simplify further).
2. $\left[\frac{5}{3}x^3 - x^2 + x \right]_1^3 = \left(45 - 9 + 3 \right) - \left(\frac{5}{3} - 1 + 1 \right) = 39 - \frac{5}{3} = \frac{112}{3} \approx 37.33$.

</details>